

SNMP Methods Across Versions

Demo reference - operations of SNMPv1, v2c and v3, segregated by the version that introduced them

SNMPv1 (1988) - The Original Five

Method	Direction	Port	What it does
GET	Manager -> Agent	161	Fetch value(s) of exact OID(s)
GETNEXT	Manager -> Agent	161	Fetch the next OID after a given one (tree traversal)
SET	Manager -> Agent	161	Write / change a value (needs read-write credential)
RESPONSE	Agent -> Manager	-	The reply carrying values or errors for any request
TRAP (v1)	Agent -> Manager	162	Unsolicited alert; unique v1-only packet format, no acknowledgement

SNMPv2c (1996) - Three New Operations (+ REPORT defined)

Method	Direction	What it does	Why it was added
GETBULK	Manager -> Agent	Fetch many rows in one request	v1 needed one round-trip per row - tables were painfully slow
INFORM	Agent -> Manager (or Mgr <-> Mgr)	An acknowledged trap - sender retries if no ACK	v1 traps are fire-and-forget; critical alerts could be lost silently
TRAPv2	Agent -> Manager	Trap reformatted as a normal PDU (sysUpTime + snmpTrapOID first)	Removed v1's oddball trap format
REPORT	Agent -> Manager	Engine / error diagnostics	Defined in v2, but barely used until v3

Also added (not methods): 64-bit Counter64 type, and per-varbind exceptions (noSuchObject, noSuchInstance, endOfMibView) instead of failing the whole request.

SNMPv3 (2002) - Zero New Operations

The demo line that lands: v3 added no methods at all. It wraps the entire v2c operation set in a security envelope (USM - User-based Security Model):

- Authentication (MD5 / SHA) - proves who sent the request and that it was not altered
- Encryption (DES / AES) - the ScopedPDU carrying OIDs and values is unreadable on the wire
- User accounts + security levels: noAuthNoPriv -> authNoPriv -> authPriv

Only behavioral addition: REPORT becomes heavily used - for engine discovery and security errors (wrong user or password -> usmStats report).

Compatibility Matrix - What Is Usable in Each Version

Method	v1	v2c	v3
GET	Yes	Yes	Yes
GETNEXT	Yes	Yes	Yes
SET	Yes	Yes	Yes
RESPONSE	Yes	Yes	Yes
GETBULK	No	Yes	Yes

TRAP	v1-format	v2-format	v2-format (secured)
INFORM	No	Yes	Yes
REPORT	No	(defined)	Yes - actively used

The 'GETWALK' Gotcha - Say This in the Demo

WALK is not a protocol method. There is no WALK PDU on the wire.

It is a manager-side loop built from the real operations:

walk in v1 = GETNEXT, GETNEXT, GETNEXT ... until end of subtree

walk in v2c / v3 = GETBULK, GETBULK ... (25+ rows per round-trip)

Same story for 'table poll' or 'device scan' - all composites of the 8 primitives above.

10-Second Demo Summary

'v1 gave us the five primitives - GET, GETNEXT, SET, RESPONSE, TRAP. v2c made it efficient: GETBULK for tables, INFORM for reliable alerts, 64-bit counters. v3 made it secure - no new methods, just authentication and encryption around the same PDUs. And walk is not a method at all - it is a loop of GETNEXT or GETBULK.'